When does a suction drain stop draining?

Two-phase hydraulics of a partly submerged perforated drain at −7 mmHg, and a hole layout that keeps it draining

USZ drain project, technical note

Analysis, simulations and text prepared with Claude Code (Anthropic)

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Abstract

A perforated drain of USZ size (elliptical body 17.85×9.67 mm, 119 mm long, closed tip, Ø4.24 mm outlet bore centred in the end face) lies in a pool of a glycerol–water mixture and is connected to a regulated suction of $p_s = -7$ mmHg (−933 Pa). As the pool empties, some holes come out of the liquid and suck air. We derive when the drainage stops, test hole layouts and sizes, and confirm the results with two independent simulations.

The air entering the exposed holes and the outlet bore act as two orifices in series. They leave a vacuum $|p_s|/(1+\Lambda^2)$ inside the drain, where Λ is the exposed open area divided by the bore area, both weighted by their discharge coefficients. Liquid can leave only through the part of the bore below the inner liquid level, so draining stops when

$$h + \frac{H_v}{1+\Lambda^2(h)} = z_\ell, \qquad H_v = \frac{|p_s|}{\rho g} = 84.5 \text{ mm},$$

where h is the pool level and z_ℓ the height of the bore's lower lip. A refined form accounts for air that must enter against the liquid column inside the drain. Across 1127 cases (11 layouts, 8 hole sizes from 0.5 to 4 mm, tilts from −10° to +20°), the refined closed form matches a two-phase network simulation with a mean difference of 0.03 mm. A time-dependent simulation reproduces the same stop levels.

The main results are:

(i) The as-built USZ field (468 mm^2 open, $\Lambda \approx 24$) stops at 2.2 mm when lying flat, but at 20.0 mm when the outlet end is 10° up and at 36.9 mm at 20°.

(ii) Hole size matters through the total open area and through how evenly the holes cover the length, not by itself.

(iii) Two staggered rows at ±30° from the bottom, 36×Ø1.0 mm (28.3 mm^2, twice the bore area), drain to the lowest hole edge, i.e. a residual level of 0.5–2.1 mm, from −10° to +20°, and pass 755 mL/min while submerged.

(iv) The intended source is a wall regulator. The recommendation holds as long as the regulator keeps −7 mmHg while passing 18 L/min of air (12 L/min if the tilt stays within 10°). A weaker source needs fewer holes; a sizing rule is given.

1 The question

The drain lies in a "lake" of liquid with a viscosity between that of water and glycerol. The suction keeps the outlet tube at −7 mmHg. Fluid enters through side holes and leaves through the outlet. The drain is not always level, and as the lake empties, part of the drain sits in air. Two questions follow:

1. For a given hole layout and hole size, at what pool level does the drain stop removing liquid?
2. Which uniform layout and hole size (0.5 to 4 mm) keeps it draining longest, so that it leaves the least liquid behind?

"Stops" means that the liquid flow through the outlet falls below 0.1 mL/min. Section 3 derives the hydraulics and a closed-form stop condition. Section 4 describes the layouts tested. Section 5 describes the two simulations used to confirm the analysis, and Section 6 gives the results. The recommended design is in Section 7.

2 Geometry, fluid and operating point

We keep the body of the USZ drain from the project package (`USZ_drain_package`) and replace its hole field (Fig. 1, Table 1). The body is an elliptic cylinder lying flat, with outer semi-axes $a = 8.925$ mm (lateral) and $b = 4.835$ mm (vertical), length $L = 119$ mm from the closed tip ($x = 0$) to the outlet end face ($x = L$), and a 0.6 mm skin. The outlet bore (Ø4.24 mm, area $A_p = 14.12$ mm^2) is centred in the end face, roughly half its 9.67 mm height. The interior is modelled as one open lumen; the fluted inner wall and core of the as-built part are not modelled (Section 8).

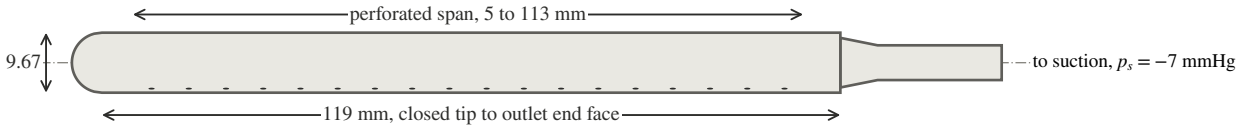
A point on the skin is located by its axial position x and by the angle θ of the parametrisation $y = a\sin\theta$, $z = b(1-\cos\theta)$, with $\theta = 0$ the bottom line (6 o'clock), ±90° the side edges and 180° the top. Holes are drilled normal to the

skin. The drain may be tilted by α about a horizontal lateral axis; $\alpha > 0$ lifts the outlet end. The pool level h is measured from the lowest point of the skin. The lower edge of the bore, the *lip*, then sits at

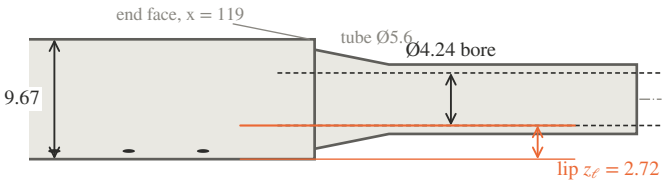
$$z_\ell(\alpha) = (b - r_p)\cos\alpha + L\max(\sin\alpha, 0), \qquad z_\ell(0) = 2.71 \text{ mm}, \tag{1}$$

which reaches 23.3 mm at $\alpha = 10°$ and 43.3 mm at 20°.

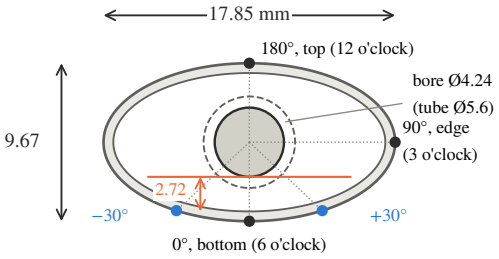
(a) Side view, lying flat, to scale (recommended hole layout on the near side)



(b) Outlet end, enlarged: centred bore and its lip



(c) End view: angle θ around the skin



(d) Tilt α (outlet up > 0), pool level h and lip height $z_\ell(\alpha)$

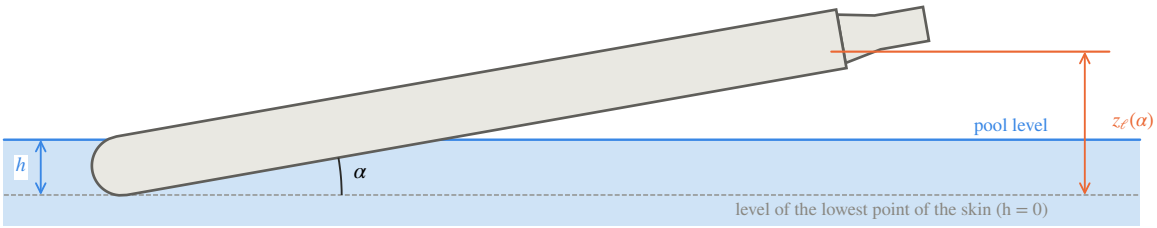


Figure 1: Drain geometry (USZ body). (a) Side view to scale, showing the holes of the recommended layout on the near side. (b) Outlet end: the Ø4.24 mm bore is centred, so its lower lip is $z_\ell = 2.72$ mm above the bottom of the skin. (c) End view and the angle θ that locates holes around the skin; the recommended rows sit at ±30° (blue). (d) Tilt α, pool level h and lip height $z_\ell(\alpha)$, Eq. (1).

Table 1: Parameters of the reference case.

quantity	value	note
outer skin	17.85×9.67 mm ellipse, $L = 119$ mm	USZ body, lying flat
skin thickness t	0.6 mm	hole length
outlet bore	Ø4.24 mm, $A_p = 14.12$ mm^2, centred	lip at $z_\ell = 2.71$ mm
perforated span	$x = 5$ to 113 mm	as in the USZ part
liquid	50 wt% glycerol–water, 20 °C	$\rho = 1126$ kg/m^3, $\mu = 6.0$ mPa s [7], $\gamma = 0.068$ N/m
air	20 °C	$\rho_a = 1.20$ kg/m^3, $\mu_a = 1.81 \times 10^{-5}$ Pa s
suction	$p_s = -7$ mmHg $= -933$ Pa	regulated in the outlet tube
suction head	$H_v = \lvert p_s\rvert/\rho g = 84.5$ mm	Eq. (2)
discharge coefficients	hole $C_h = 0.61$ ($d \geq 0.8$ mm) to 0.68 ($d = 0.5$ mm), bore $C_p = 0.85$	Eqs. (4), (8)
pool footprint	200×80 mm	only for drain times

3 Physics

All pressures are gauge pressures; the atmosphere is $p = 0$.

3.1 Suction head

At −7 mmHg the suction can hold up a liquid column of

$$H_v = \frac{|p_s|}{\rho g} = \frac{933.3}{1126 \times 9.81}\ \text{m} = 84.5\ \text{mm}. \tag{2}$$

This is about nine times the drain height. Gravity and capillarity are therefore not negligible next to the suction, and they decide when the drain stops.

3.2 Flow through one hole

A hole of diameter d through a wall of thickness t resists flow through a viscous term and an inertial term in series:

$$\Delta p = R_v\, q + K\, q|q|, \qquad R_v = \mu\left(\frac{24}{d^3} + \frac{128\,t}{\pi d^4}\right), \qquad K = \frac{\rho}{2C_h^2 A^2}, \qquad A = \frac{\pi d^2}{4}. \tag{3}$$

The first viscous part is Sampson's creeping flow through a circular aperture in a thin plate [1]. The second is Hagen–Poiseuille flow along the bore. Their sum approximates the exact finite-length solution well [2]. The inertial part is the orifice law with discharge coefficient C_h, which depends on t/d [3]:

$$C_h = 0.61 \ \ (t/d \le 0.75), \qquad C_h \text{ rising linearly to } 0.81 \ \ (t/d \ge 2). \tag{4}$$

For $t = 0.6$ mm this gives $C_h = 0.61$ for $d \ge 0.8$ mm. Solving Eq. (3) for q gives the numerically stable form

$$q = \frac{2\,\Delta p}{R_v + \sqrt{R_v^2 + 4K\Delta p}}. \tag{5}$$

For the liquid at $\Delta p \approx 0.5$ kPa and $d = 1$ mm the hole Reynolds number is about 100, and the viscous term lowers the flow by about 12 % relative to the pure orifice law; at $d = 0.5$ mm it lowers it by about 40 %. For air, $\mu_a/\mu = 0.003$ and the viscous term is negligible. A hole that is partly submerged is split by area into a submerged fraction f_s and an exposed fraction f_e, each treated as a hole of area fA (that is, R_v/f and K/f^2).

3.3 Why an exposed hole leaks air

When the liquid surface drops below a hole, a meniscus forms in it. It can resist the suction behind it only up to the capillary entry pressure

$$P_c = \frac{4\gamma\cos\theta_e}{d} \le \frac{4\gamma}{d}, \tag{6}$$

where θ_e is the effective contact angle (with edge pinning the maximum is $4\gamma/d$ [5, 6]). The largest possible value is 544 Pa for $d = 0.5$ mm, 272 Pa for $d = 1$ mm and 68 Pa for $d = 4$ mm. All of these are below $|p_s| = 933$ Pa. A hole seals against the full suction only if

$$d < d^* = \frac{4\gamma}{|p_s|} = 0.29\ \text{mm}, \tag{7}$$

even with perfect wetting. Every hole in the 0.5 to 4 mm range therefore lets air in as soon as it is exposed. Once air passes, the hole stays open until it is submerged again. Capillarity plays a role only after the vacuum inside the drain has already dropped, and it is treated as a sensitivity case ($\theta_e = 60°$ and $0°$; the baseline is $\theta_e = 90°$, no holding).

3.4 The outlet bore

The flow contracts from the drain lumen ($\approx$ 110 mm^2) into the bore (14.1 mm^2), an area ratio $\sigma \approx 0.13$. Weisbach's contraction coefficient $C_c = 0.63 + 0.37\sigma^3 \approx 0.63$ gives a loss coefficient $K_c = (1/C_c - 1)^2 \approx 0.35$ [4]. Because p_s is the static pressure in the outlet tube, the bore behaves as an orifice with

$$C_p = (1 + K_c)^{-1/2} \approx 0.85, \qquad Q_{max} = C_p A_p \sqrt{\frac{2|p_s|}{\rho}} = 927\ \text{mL/min}. \tag{8}$$

Q_{max} is the most liquid that the bore can pass at −7 mmHg. As Section 6 shows, the bore, not the holes, is the main restriction whenever the open hole area exceeds about twice the bore area.

3.5 The flow network at one pool level

Inside the drain there is a gas pressure p_g and a horizontal inner liquid level ζ; the drain is full of liquid when ζ lies above its top. The pressure on the inner side of a hole at height z is

$$p_{in}(z) = p_g + \rho g\,(\zeta - z)^+, \tag{9}$$

where $(\cdot)^+$ means $\max(\cdot, 0)$. For opening j, with submerged and exposed parts at mean heights z_j^s and z_j^e, the network has the following flows:

$$\text{liquid in:} \quad \Delta p_j^L = \rho g\,(h - z_j^s) - p_{in}(z_j^s), \tag{10}$$

$$\text{air in:} \quad \Delta p_j^G = -p_{in}(z_j^e) \ \text{ if the opening is open to air,} \tag{11}$$

$$\text{liquid out of an exposed hole:} \quad \Delta p = p_{in}(z_j^e) > 0 \ \text{ if its inner side is wet.} \tag{12}$$

Each flow follows Eq. (5). An exposed opening becomes open to air when $-p_{in}(z_j^e) > P_c$. At the bore, gravity separates the phases (a separated-flow model [8]): liquid leaves through the part of the bore below ζ and gas through the part above it,

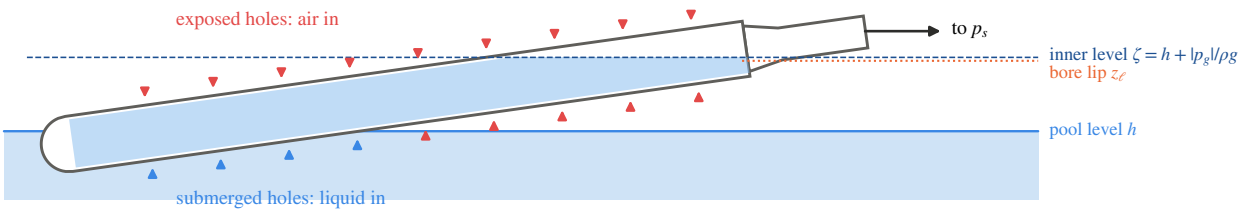
$$A_L = A_p\,F\!\left(\frac{\zeta - z_c}{r_p \cos\alpha}\right), \quad F(u) = \frac{\pi - \arccos u + u\sqrt{1-u^2}}{\pi}, \qquad A_G = A_p - A_L, \tag{13}$$

$$q_{L,b} = C_p A_L \sqrt{\frac{2\,(p_g - p_s + \rho g\,(\zeta - \bar{z}_L))}{\rho}}, \qquad q_{G,b} = C_p A_G \sqrt{\frac{2\,(p_g - p_s)}{\rho_a}}, \tag{14}$$

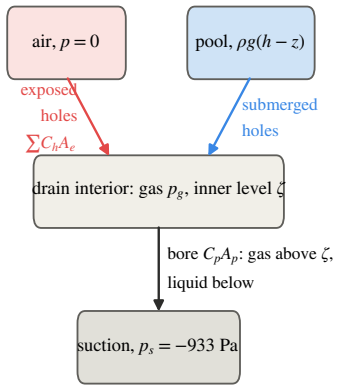
where z_c is the bore centre and $\bar{z}_L$ the centroid of the liquid part. The two unknowns (p_g, ζ) follow from conservation of liquid and gas:

$$\sum_j q_j^L - \sum_j q_j^{leak} = q_{L,b}, \qquad \sum_j q_j^G = q_{G,b}. \tag{15}$$

(a) Partly drained, tilted drain: air entering the exposed holes lowers the vacuum p_g inside



(b) Flow network at one pool level



(c) Lift left by the pressure divider

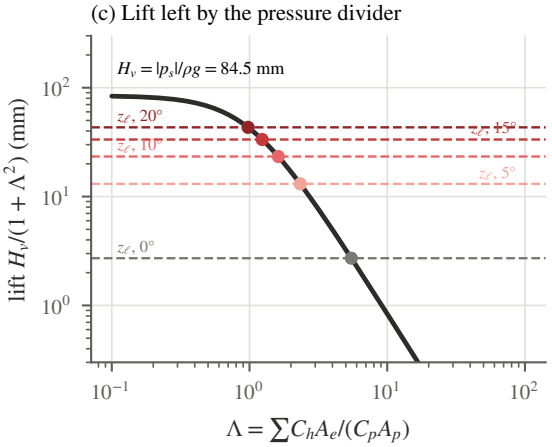


Figure 2: Mechanism. (a) Once holes are exposed, air enters through them while liquid still enters through the submerged ones. The liquid inside the drain stands at $\zeta = h + |p_g|/\rho g$ and can leave only through the part of the bore below ζ. (b) The flow network solved at each pool level, Eqs. (10)–(15). (c) The vacuum left inside the drain, expressed as a liquid lift, against the exposed-area ratio Λ. The dashed lines are the lip heights $z_\ell(\alpha)$, and the dots mark the largest Λ that still lifts the liquid to the lip.

3.6 The pressure divider

Neglect the liquid column inside the drain and the viscous gas terms. The gas balance in Eq. (15) then involves two orifices in series, the exposed holes and the bore:

$$\sum_j C_{h,j} A_j^e \sqrt{\frac{2(-p_g)}{\rho_a}} = C_p A_p \sqrt{\frac{2(p_g - p_s)}{\rho_a}} \quad \Longrightarrow \quad -p_g = \frac{|p_s|}{1+\Lambda^2}, \qquad \Lambda = \frac{\sum_j C_{h,j} A_j^e}{C_p A_p}.$$

The gas density and the size of the air flow both drop out, so the vacuum left inside depends only on the area ratio Λ. With $\Lambda = 1$, half the suction remains. The as-built USZ field reaches $\Lambda \approx 12$ as soon as its top face is exposed, which leaves 0.7 % of the suction, or 0.6 mm of lift.

3.7 Lift, and the stop condition

Liquid can leave only if the inner level reaches the bore lip, $\zeta \geq z_\ell$. When no liquid flows, the pressure balance across a submerged hole is $\rho g(h - z) = p_g + \rho g(\zeta - z)$, so the vacuum lifts the inner liquid to

$$\zeta = h + \frac{|p_g|}{\rho g}. \tag{16}$$

Drainage is therefore possible only while $h + |p_g|/\rho g > z_\ell$. Combined with the pressure divider, the drain stops at the level h_s that satisfies

$$h_s + \frac{H_v}{1 + \Lambda(h_s)^2} = z_\ell \qquad \text{(simple bound)}. \tag{17}$$

This bound treats every exposed hole as if air entered it at the full inner vacuum. An exposed hole that lies *below* the inner level ($z < \zeta$) admits air only by pushing against the liquid column $\rho g(\zeta - z)$. At the threshold, where $\zeta = z_\ell$ and $-p_g = \rho g(z_\ell - h)$, the air entering an exposed element at height z is driven by $\rho g(\min(z, z_\ell) - h)$ instead of $\rho g(z_\ell - h)$. Weighting each exposed element by $w = (\min(z, z_\ell) - h)/(z_\ell - h)$ gives the refined stop condition

$$\boxed{h_s + \frac{H_v}{1 + \Lambda_w(h_s)^2} = z_\ell, \qquad \Lambda_w(h) = \frac{1}{C_p A_p} \sum_{\text{exposed}} C_h \, \delta A \, \sqrt{w}} \qquad \Lambda_w \leq \Lambda. \tag{18}$$

The left side increases with h, because Λ_w never increases as h rises, so the root is unique. Drainage also needs liquid that can reach a hole, so it stops at the latest when the pool falls below the lowest edge of the lowest hole, h_{geo}. Overall, $h_s = \max(h_{geo},$ root of Eq. (18)). Three consequences follow:

1. While $h > z_\ell$, no lift is needed: liquid reaches the bore whatever the vacuum, so the drain keeps draining. Every drain therefore empties the pool at least down to the lip, $h_s < z_\ell$, which is 2.7 mm for a drain lying flat.
2. With the outlet up, the lip rises by $L \sin\alpha$, and the vacuum has to lift the liquid much further.
3. The drain empties the pool down to its lowest hole only if

$$\Lambda_w(h_{geo}) < \Lambda^* = \sqrt{\frac{H_v}{z_\ell(\alpha) - h_{geo}} - 1}. \tag{19}$$

Taking $h_{geo} = 0$ gives $\Lambda^* = 5.5$ at 0°, 2.3 at 5°, 1.6 at 10° and 1.0 at 20°. Beyond $\alpha \approx 44°$ the lip is higher than H_v, and no layout can empty the pool below $z_\ell - H_v$.

3.8 A suction source with limited capacity

A real pump or regulator holds p_s only up to some volume flow Q_s. Once the air drawn through the exposed holes exceeds Q_s, the source is saturated, and the vacuum inside follows from the holes alone, $|p_g| \approx (\rho_a/2)\,(Q_s/\sum C_h A^e)^2$. Full drainage then additionally requires

$$\sum C_h A^e \; < \; \frac{Q_s}{\sqrt{2\rho g\,(z_\ell - h_{geo})/\rho_a}}. \tag{20}$$

For example, at $\alpha = 10°$ a 5 L/min source can feed only about 7 mm^2 of exposed hole area at the vacuum that the lift requires.

3.9 Draining in time

The pool level falls as $\mathrm{d}h/\mathrm{d}t = -Q_L(h)/A_{fs}(h)$, where Q_L is the net liquid inflow to the drain and A_{fs} is the pool's free-surface area (pool footprint minus the drain's footprint at that level). Because the drain interior settles within seconds, much faster than the pool, $Q_L(h)$ can be taken from the steady network at each level (Section 5). Air entrainment by surface drawdown above an active hole (critical submergence [9]) is included as a sensitivity case: a hole is counted as exposed once the surface is within κd of it, with $\kappa = 0.5$.

4 Hole distributions tested

Ten uniform layouts (A to J) plus the as-built USZ field were tested (Figs. 3, 5). In every layout the holes are equally spaced along x within each row. The layouts differ in the number of rows, their angles θ and their axial alignment (in line, staggered by half a pitch, or helical):
A side rows (3 and 9 o'clock); B top and bottom; C four rows in line; D helix over the same four positions; E eight rows evenly around the skin; F both faces at ±45° and ±135°; G lower band (0°, ±45°); H lower flanks ±45° in line; I the same staggered; J one bottom row. The recommended layout (two staggered rows at ±30°) came out of the optimisation in Section 6.3.

Each layout was built with eight hole diameters, $d = 0.5$ to 4.0 mm in 0.5 mm steps, in two families (Fig. 4):
Fixed count: 19 holes per row (pitch 5.7 mm), so the open area grows as d^2.
Fixed area: the number of holes is chosen so that the total open area is about $2A_p = 28\ \mathrm{mm}^2$, so larger holes mean fewer holes.
Tilts of −10° to +20° in 5° steps give 1127 cases.

5 Simulations used to confirm the analysis

Three-dimensional two-phase CFD (volume of fluid) was not available on this machine. The closed form was instead checked against two numerical models that drop its simplifications.

Quasi-static two-phase network. Each opening is split into 61 equal-area elements (a Vogel spiral on the developed skin), mapped to their true height for the tilt. At each pool level the solver finds the submerged and exposed fractions of every opening and solves Eqs. (10)–(15). It includes:

- viscous and inertial losses in every opening;
- hydrostatics inside the drain, and liquid dripping out of exposed holes;
- gas and liquid sharing the bore;
- the capillary breakthrough state of each opening, which persists until the opening is submerged again;
- optionally, a source with limited capacity.

The two unknowns are found by nested monotone root finding (Illinois false position): p_g from the gas balance for a given ζ, and ζ from the liquid balance. If the liquid balance is negative even with ζ at the lip, the drain is stalled. The pool is lowered in 0.1 mm steps, and the stop level is then refined by bisection to 0.01 mm. Drain times come from integrating $\mathrm{d}t = A_{fs}\,\mathrm{d}h/Q_L$.

Time-dependent simulation. This model does not assume a steady interior. The liquid stored inside the drain, V_L, is a state variable:

$$\frac{\mathrm{d}V_L}{\mathrm{d}t} = Q_{L,in} - Q_{leak} - q_{L,b}, \qquad \frac{\mathrm{d}h}{\mathrm{d}t} = -\frac{Q_{L,in} - Q_{leak}}{A_{fs}}, \qquad 0 = Q_{L,in} + Q_{G,in} - Q_{leak} - q_{L,b} - q_{G,b}. \tag{21}$$

The last equation states that the rigid drain conserves volume, and it fixes p_g at every instant (gas compressibility acts on microsecond scales at 1 % pressure changes). The inner level ζ follows from V_L through the interior geometry. V_L is advanced with backward Euler, which is stable for the stiff bore, and h explicitly with steps of at most 0.05 mm. Openings break through and reseal as in the network model. The run ends once no liquid has left the drain for two minutes.

Agreement. Over all 1127 sweep cases, the refined closed form, Eq. (18), matches the network simulation with a mean absolute difference of 0.03 mm and a maximum of 1.27 mm (Fig. 6). The simple bound, Eq. (17), is conservative: it overestimates the stop level by up to 9.4 mm at outlet-up tilts. The time-dependent simulation ends at the same levels to within 0.02 mm (Table 2); Fig. 7 shows its time histories.

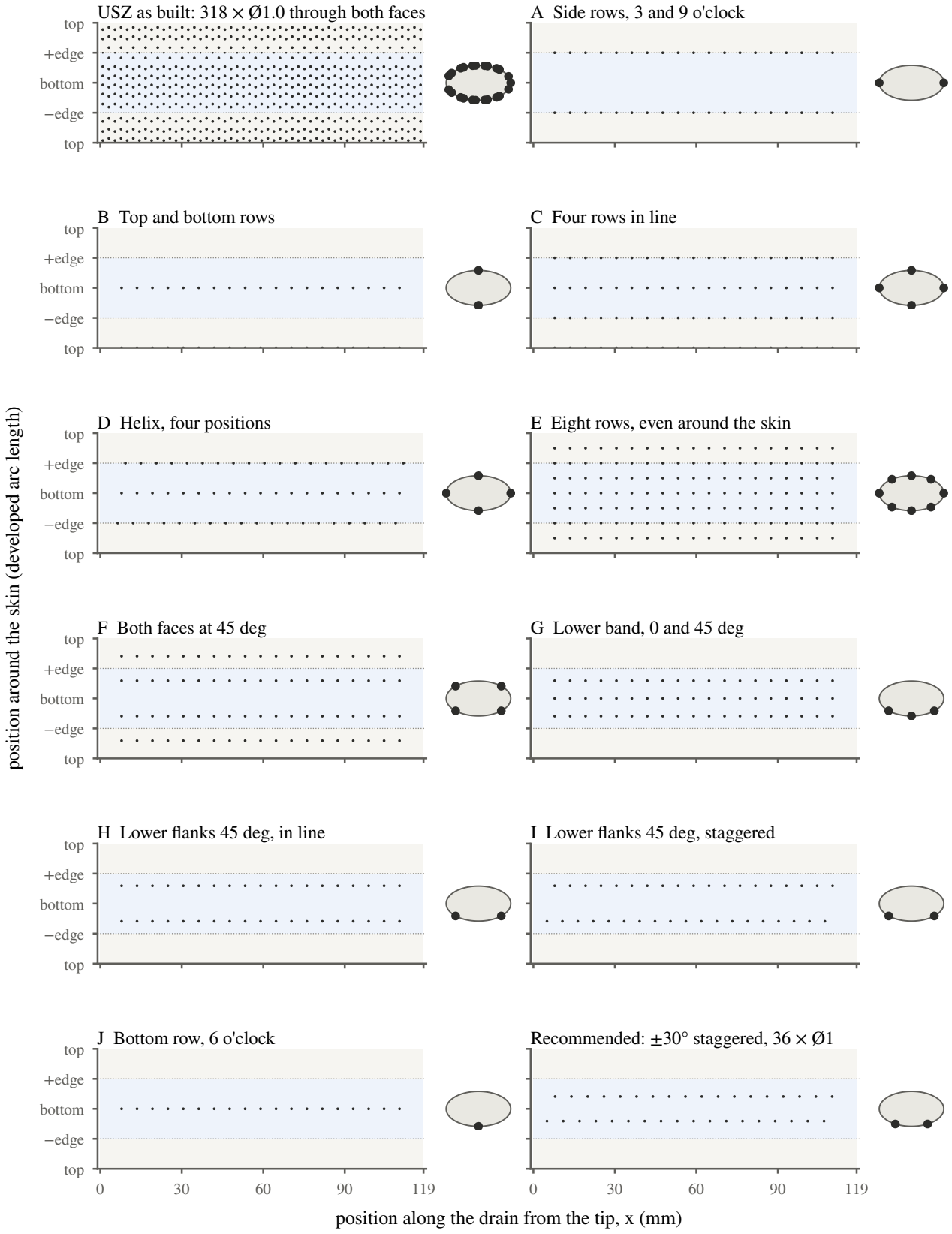


Figure 3: All tested hole distributions, drawn on the unrolled skin (to scale, Ø1.0 mm, 19 holes per row). The horizontal axis is the position along the drain; the vertical axis runs once around the skin (bottom in the middle, side edges dotted, top at both edges). The small end views show where the rows sit. Dark holes on the light-blue band face downward when the drain lies flat. The recommended layout (last panel) has two rows at ±30°, staggered by half a pitch.

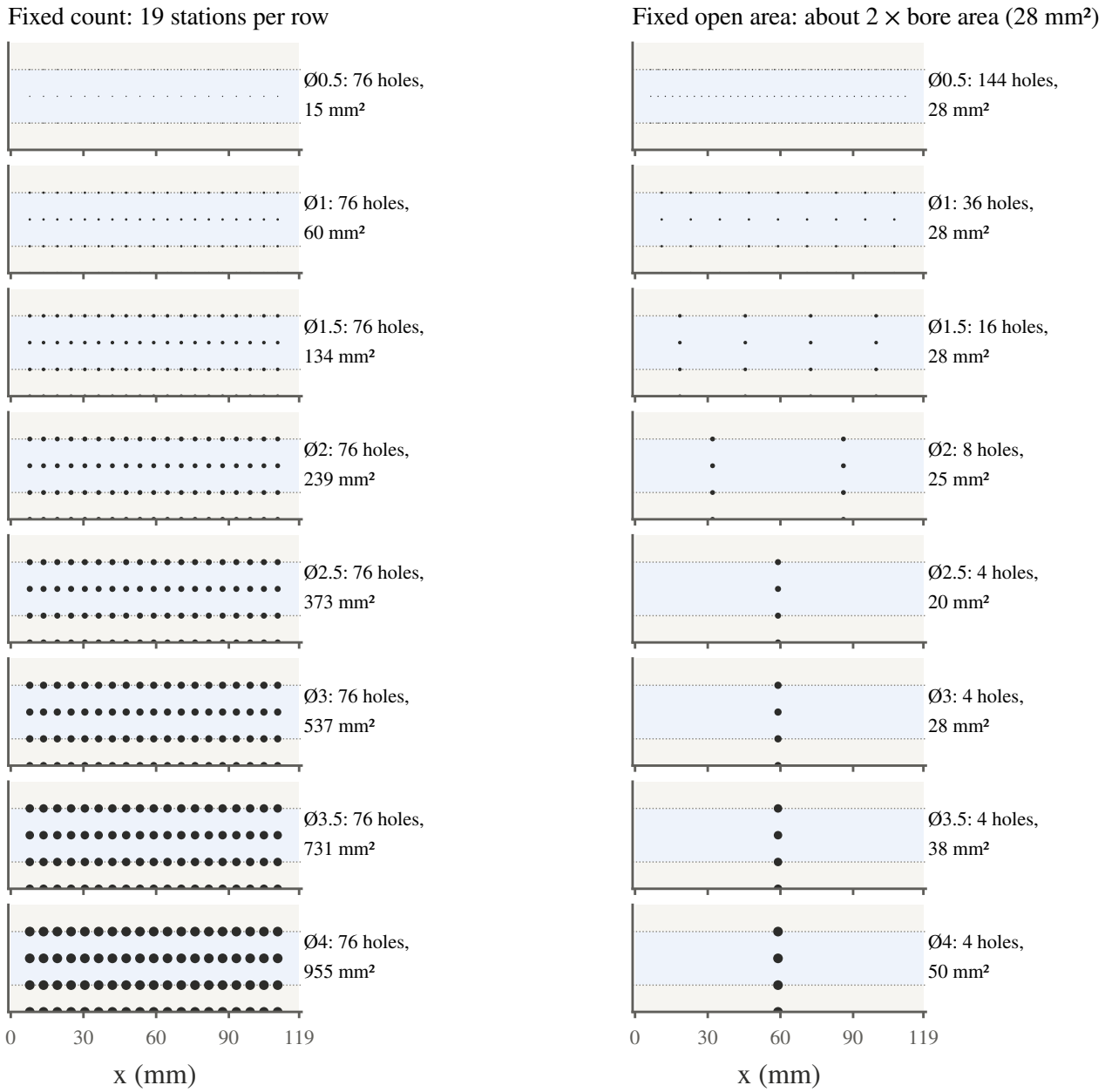


Figure 4: Hole-size families, shown for layout C. Left: fixed count, so the open area grows from 15 to 955 mm^2. Right: fixed open area of about 28 mm^2. Large holes then come in very few stations; from 2.5 mm on, all holes sit at mid-length, which leaves the ends of the drain without holes.

Table 2: Stop level h_s (mm above the lowest point of the skin) from the three methods, for four designs. "Four rows" and "side rows" are the fixed-area (28 mm^2) versions.

design	α	closed form Eq. (18)	network (quasi-static)	time-dependent simulation	lowest hole edge h_{geo}
Recommended	0°	0.51	0.52	0.51	0.53
Recommended	10°	1.35	1.36	1.36	1.37
USZ as built	0°	2.29	2.21	2.21	0.01
USZ as built	10°	20.13	19.97	19.97	0.11
Side rows	0°	4.35	4.37	4.36	4.37
Side rows	10°	5.67	5.67	5.67	5.69
Four rows	0°	0.00	0.00	0.00	0.01
Four rows	10°	1.81	1.83	1.82	1.83

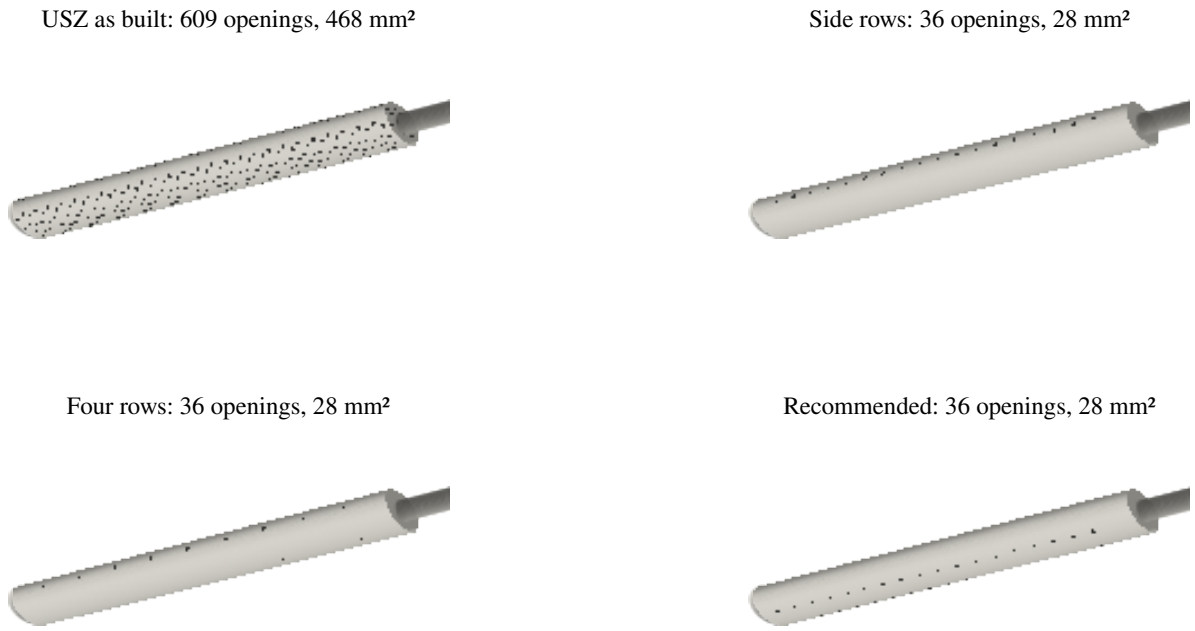


Figure 5: 3D views (from below and in front, outlet tube on the right) of the as-built USZ field, side rows, four rows and the recommended layout.

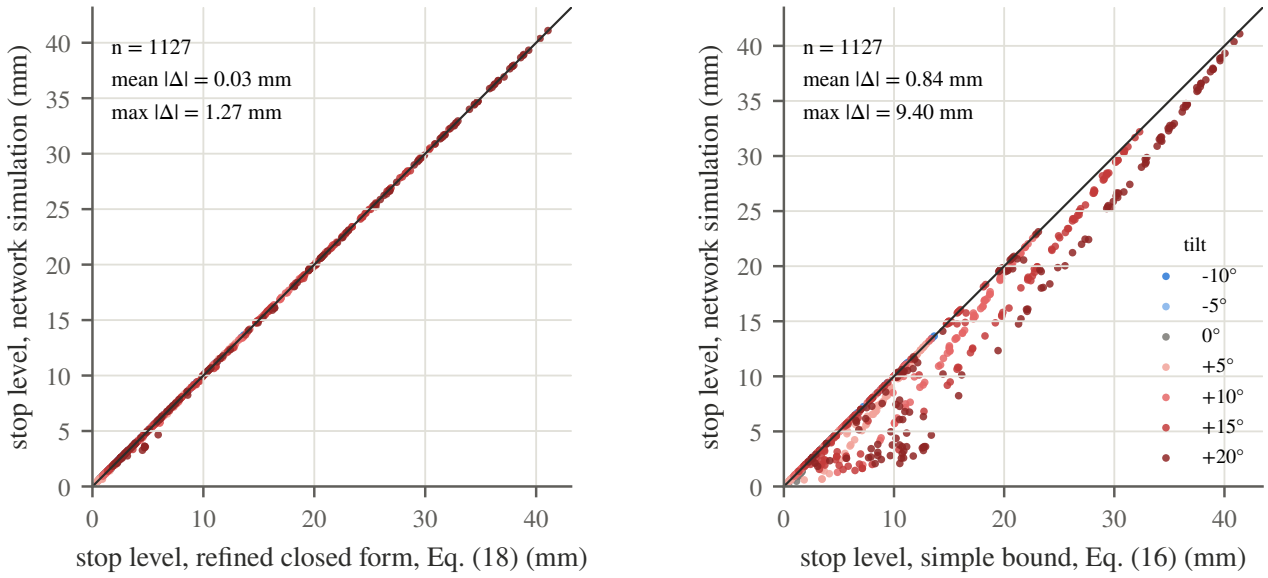


Figure 6: Stop level from the network simulation against the closed forms, for all 1127 cases (colour gives the tilt). Left: refined condition, Eq. (18). Right: simple bound, Eq. (17), which is conservative when air must enter against liquid standing inside a tilted drain.

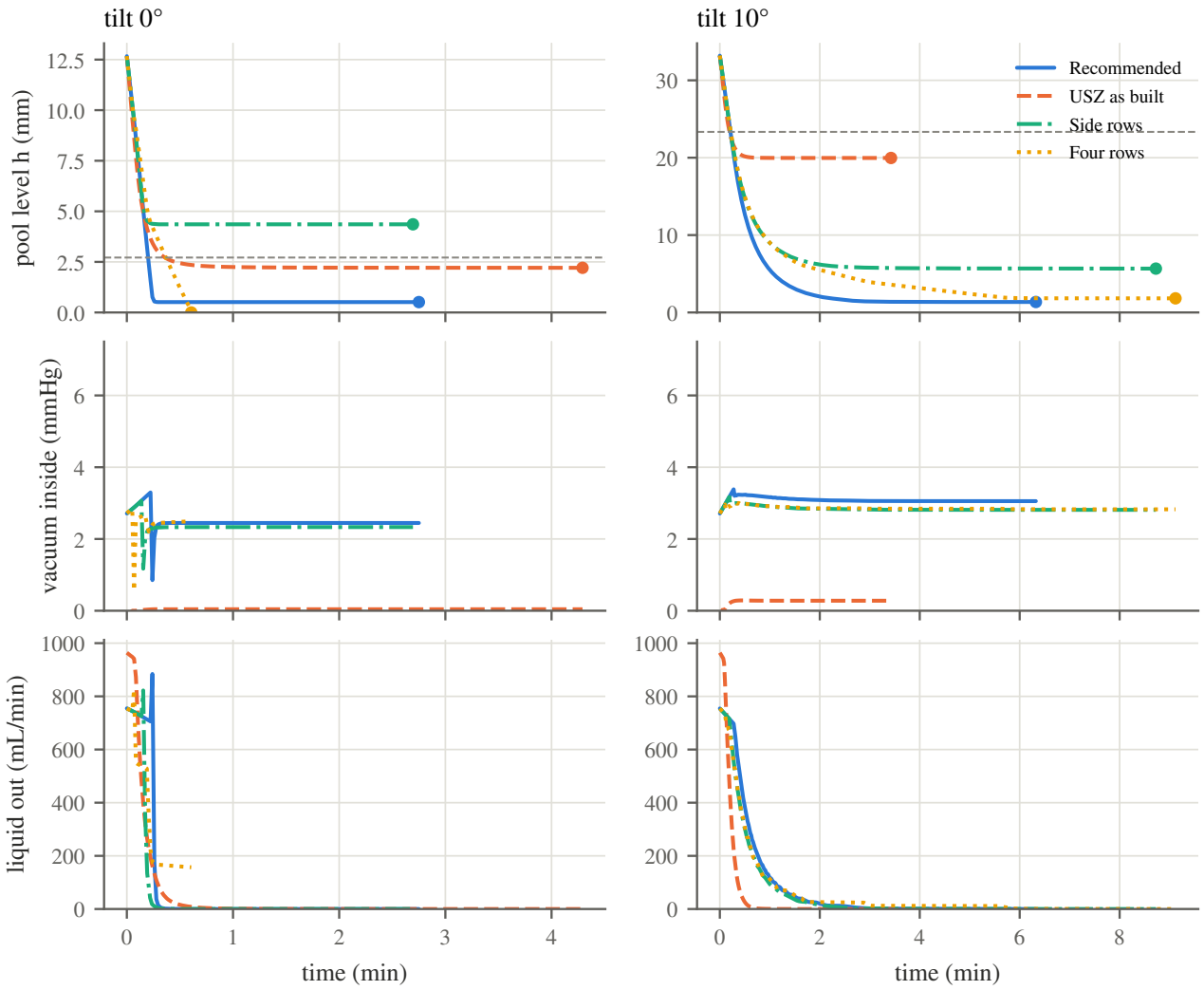


Figure 7: Time-dependent simulation (pool 200 × 80 mm, starting 3 mm above the drain): pool level, vacuum inside the drain and liquid leaving through the bore. The dots mark where the drainage has stopped.

6 Results

6.1 How the drainage stops

Figure 8 follows five designs as the pool falls. While all holes are submerged, the drains pass 720 to 970 mL/min, close to the bore limit Q_{max} = 927 mL/min. The vacuum at the bore end of the interior is only about 2.4 mmHg; the bore itself takes the rest of the 7 mmHg. With the large USZ field, the interior even sits slightly above atmospheric pressure, because all the suction acts across the bore.

As soon as the upper holes of the USZ field are exposed, the interior vacuum collapses ($\Lambda \gg 1$). The drain then keeps emptying only while the pool is above the lip, which is 2.7 mm when flat and 23.3 mm at 10°, and it stops just below. A layout with little open area keeps 2 to 3 mmHg inside and drains on until its lowest holes are uncovered. Side rows stop at about mid-height (4.4 mm when flat): this is a geometric stop, because nothing lower can take in liquid.

6.2 Effect of layout, hole size and tilt

Table 3 and Figs. 9–11 give the stop level of every case. The main findings:

- **The as-built USZ field** drains well only when flat, stopping at 2.2 mm just below the lip. Any outlet-up tilt leaves a deep pool (20.0 mm at 10°), because its 468 mm^2 of openings give $\Lambda \approx 24$.
- **Total open area is the main design variable.** With a fixed count (open area $\propto d^2$), the layouts that tolerate tilt with small holes fail with the outlet up once d exceeds about 1 to 2 mm, when the open area passes roughly 60 to 130 mm^2 (Fig. 9). With a fixed area of 28 mm^2, every layout stops below about 7 mm at all tilts, and the side rows are the worst when flat (Table 3).
- **Placement around the skin** sets the geometric limit. Holes at mid-height or above (A, and the upper parts of E and F) cannot remove the last millimetres, while holes on the lower half can.

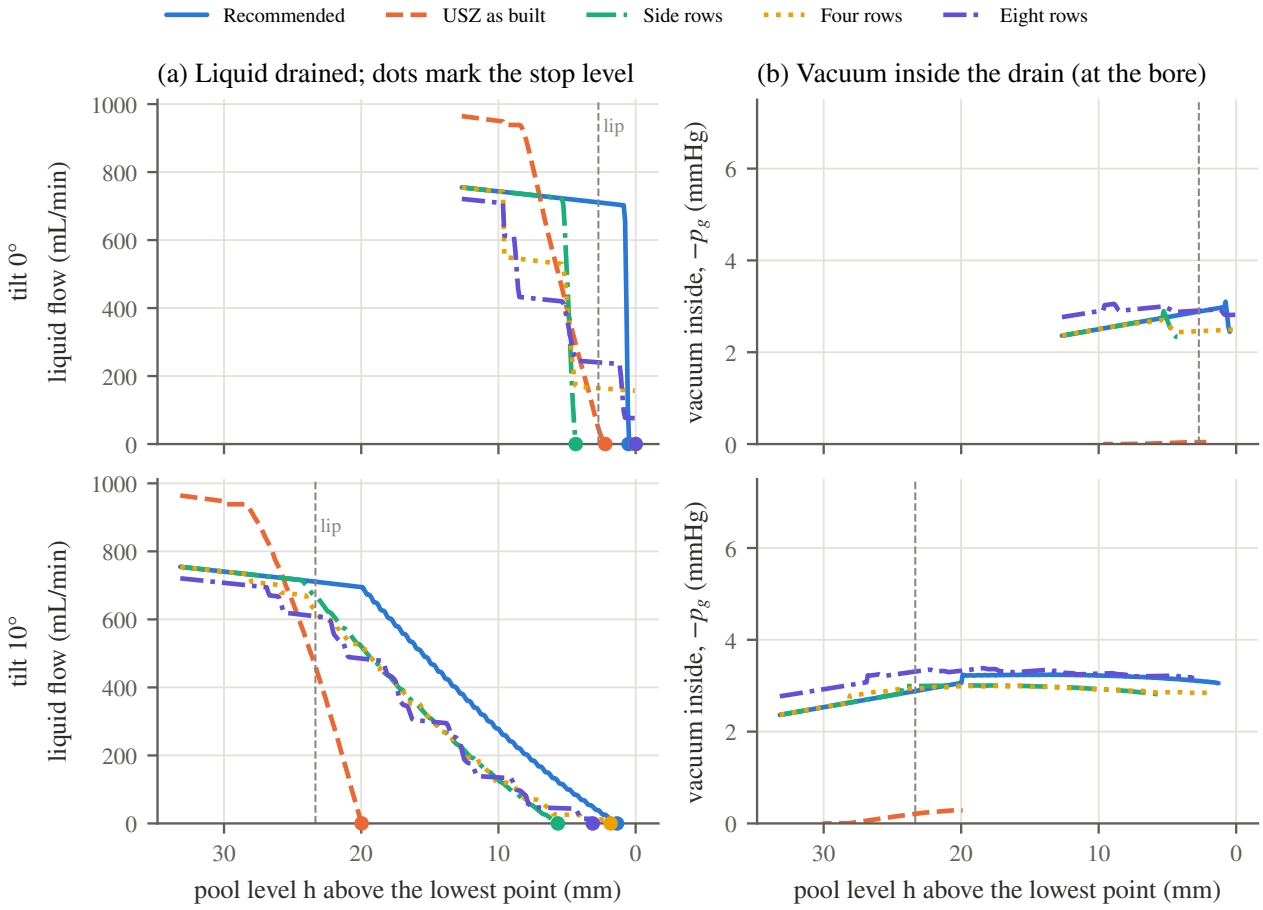


Figure 8: Quasi-static drainage of five designs, read from right to left as the pool falls. (a) Liquid drained; the dots mark the stop level. (b) Vacuum inside the drain at the bore. The dashed line is the lip height $z_\ell(\alpha)$. Top row: flat; bottom row: outlet 10° up.

- **Placement along the length** matters under tilt. The lowest point is the tip when the outlet is up and the outlet end when it is down, so holes must reach both ends. With a fixed area, large holes come in so few stations that the ends stay uncovered (Fig. 10, Table 4). Staggering two rows halves the effective axial pitch, which is why layout I beats H at positive tilt.

Table 3: Stop level h_s (mm) of all layouts with Ø1.0 mm holes at about 28 mm^2 open area (USZ as built), from the network simulation. Q_0 is the flow with the pool 3 mm above the drain.

			stop level h_s (mm) at tilt α						
layout	A (mm^2)	Q_0 (mL/min)	−10°	−5°	0°	+5°	+10°	+15°	+20°
USZ as built (318 × Ø1.0)	468	965	2.5	2.5	2.2	11.6	20.0	28.5	36.9
A Side rows, 3 and 9 o'clock	28	755	5.9	5.1	4.4	5.0	5.7	6.3	6.8
B Top and bottom rows	28	755	1.5	0.7	0.0	0.7	1.3	1.9	2.6
C Four rows in line	28	755	2.0	1.0	0.0	0.9	1.8	2.7	3.6
D Helix, four positions	28	755	2.0	1.0	0.0	0.9	1.8	2.7	3.6
E Eight rows, even around the skin	25	721	3.3	1.7	0.0	1.6	3.1	4.7	6.2
F Both faces at 45°	28	755	3.3	2.2	1.2	2.1	3.1	4.0	4.8
G Lower band, 0 and 45°	28	755	1.7	0.9	0.0	0.8	1.6	2.3	3.1
H Lower flanks 45°, in line	28	755	2.7	2.0	1.2	1.9	2.6	3.2	3.8
I Lower flanks 45°, staggered	28	755	2.7	2.0	1.2	1.6	2.0	2.4	2.8
J Bottom row, 6 o'clock	28	755	1.2	0.6	0.0	0.5	1.0	1.6	2.1
Recommended, ±30° staggered	28	755	2.0	1.3	0.5	0.9	1.4	1.8	2.1

6.3 Optimising the layout

Two staggered rows were then mapped over hole size (0.5 to 4 mm) and number of holes (Fig. 12). Flow when submerged rises with open area and reaches 80 % of Q_{max} at about $2A_p$ (Fig. 12a, Fig. 13b). The worst stop level over −10° to +15° has a broad minimum between about 15 and 40 mm^2 (Fig. 12b):

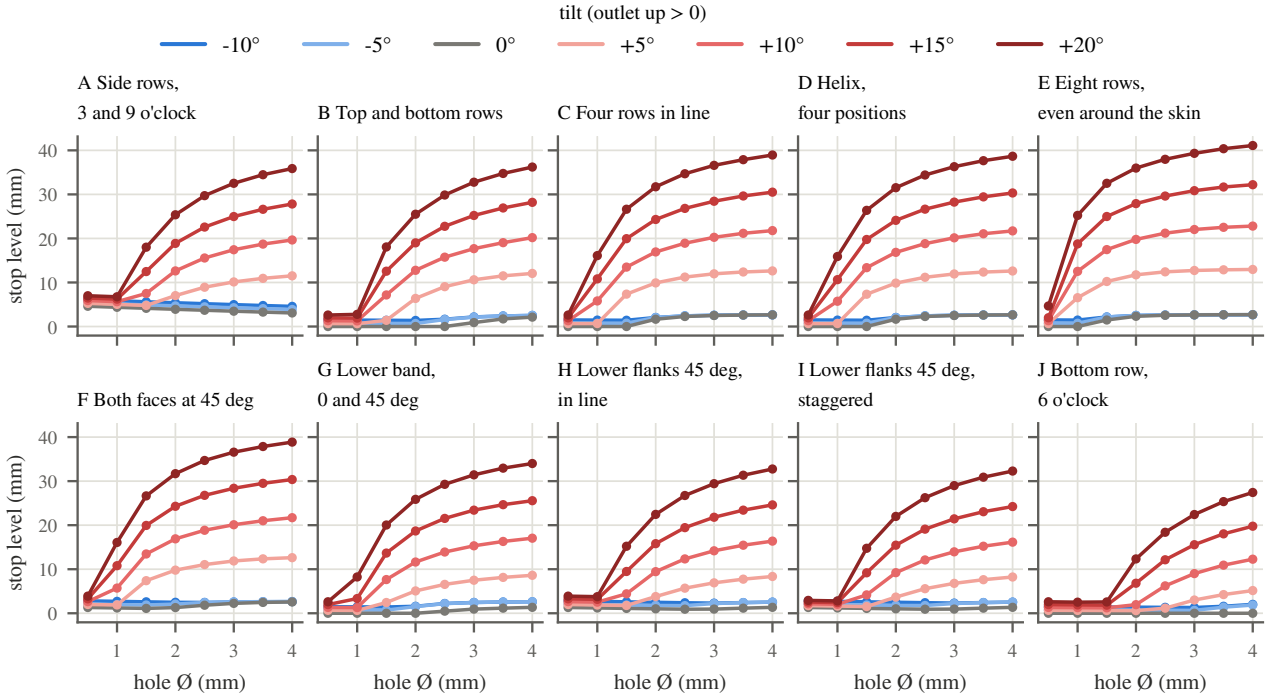


Figure 9: Stop level against hole diameter, fixed-count family (19 holes per row). Colour gives the tilt: blue for outlet down, grey for flat, red for outlet up.

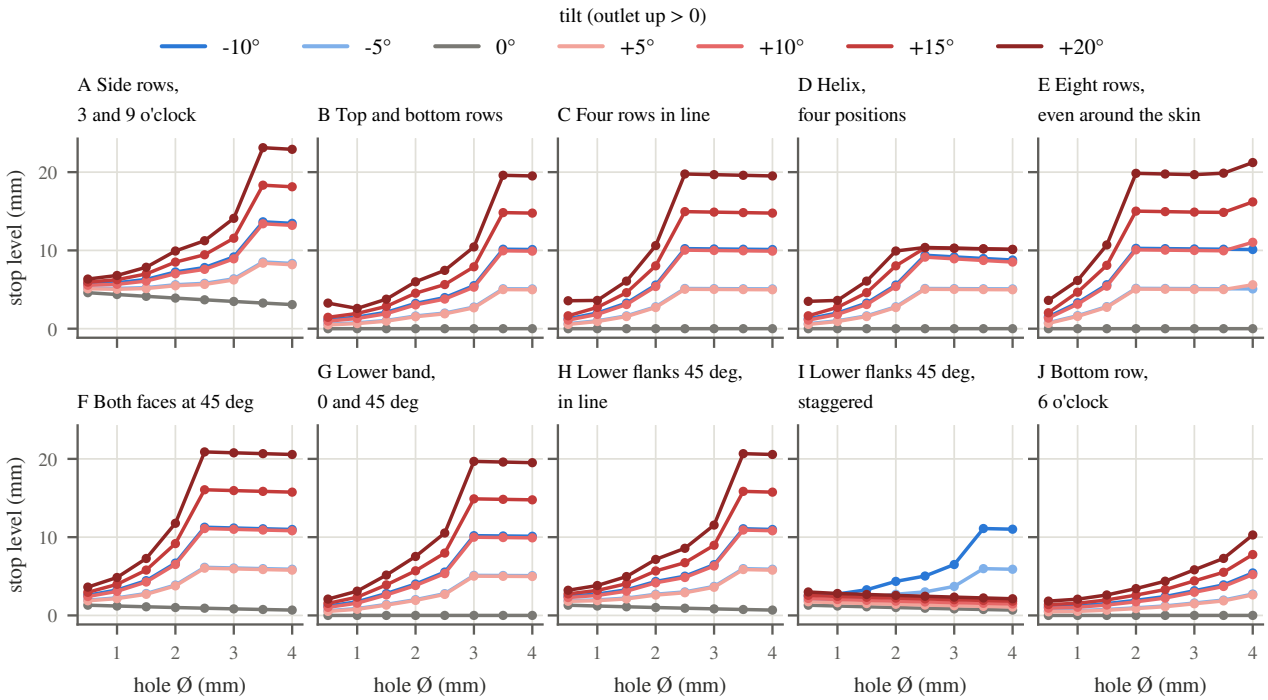


Figure 10: Stop level against hole diameter, fixed-area family (about 28 mm^2). Larger holes mean fewer stations; above about 2.5 mm the holes no longer reach the ends of the drain.

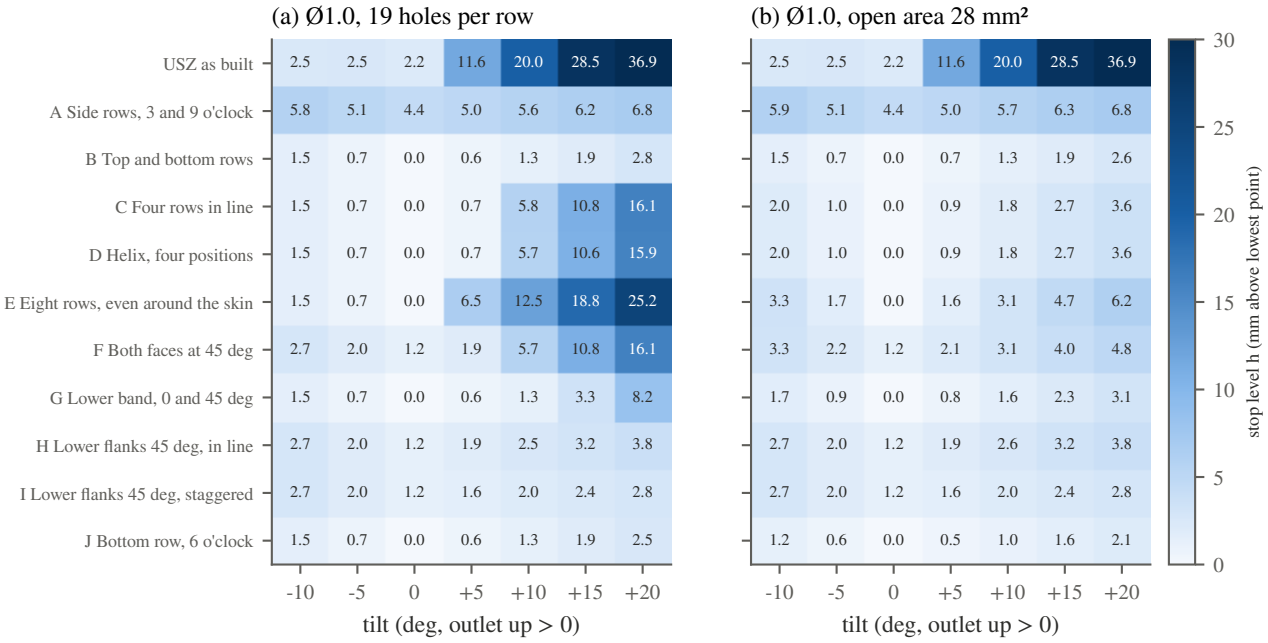


Figure 11: Stop level (mm) for every layout with Ø1.0 mm holes and tilts from −10° to +20°. (a) 19 holes per row. (b) Open area of about 28 mm^2.

- With less area there are too few holes to reach the ends.
- With more area, Λ_w exceeds Λ^* (Eq. (19)), and the lift fails at outlet-up tilts.

The rows should sit as low as practical (Fig. 12c): every 15° lower takes 0.5 to 1 mm off the residual level. However, rows below about ±20° press against a floor or against tissue (when lying on a hard floor, the holes at ±15° lose 70 % of their flow). We therefore chose ±30°, roughly 5 and 7 o'clock. At a fixed area of 28 mm^2 (Table 4), holes of 1.0 to 1.5 mm give the best balance:

- 0.5 mm holes lose flow to viscosity and clog easily.
- 2 to 4 mm holes come in too few stations and leave the outlet end uncovered when it is lowest.

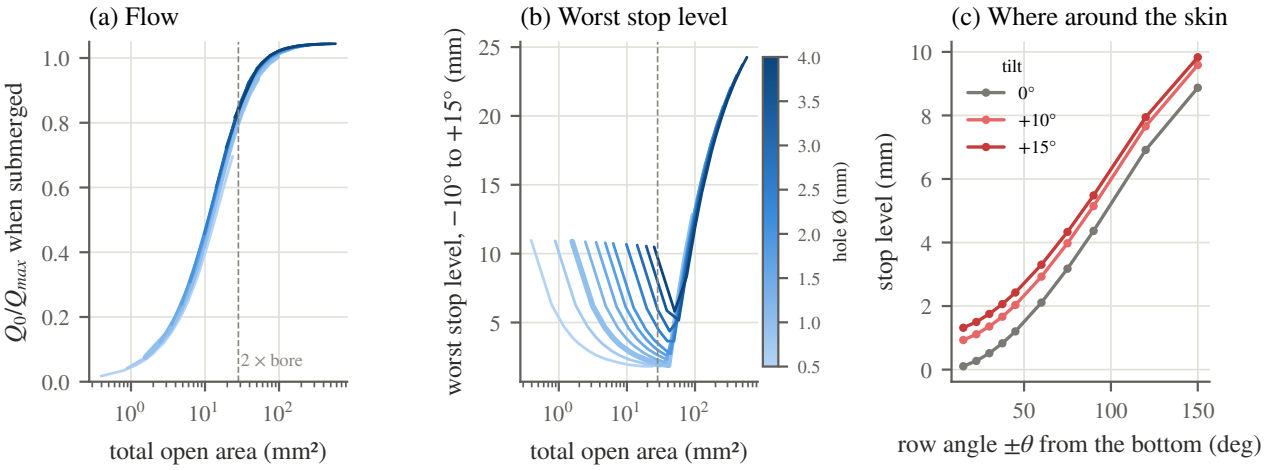


Figure 12: Optimisation of two staggered rows at ±30°. Each line is one hole size, with the number of holes varied along it. (a) Flow when submerged as a fraction of the bore limit. (b) Worst stop level over −10° to +15° (closed form). The ring marks the recommended design, and the dashed line is twice the bore area. (c) Effect of the row angle ±θ (Ø1.0 mm, 28 holes).

6.4 How far can the drain be tilted?

Figure 14 extends the tilt to 40°. With the outlet down, all layouts stop within about 3 mm of the bottom (side rows excepted). With the outlet up, the recommended design drains to its lowest hole edge up to $\alpha = 22.5°$. Beyond that, z_ℓ grows faster than the vacuum can lift, and the stop level rises by about 1 mm per degree. The as-built USZ field fails from the first degree of outlet-up tilt.

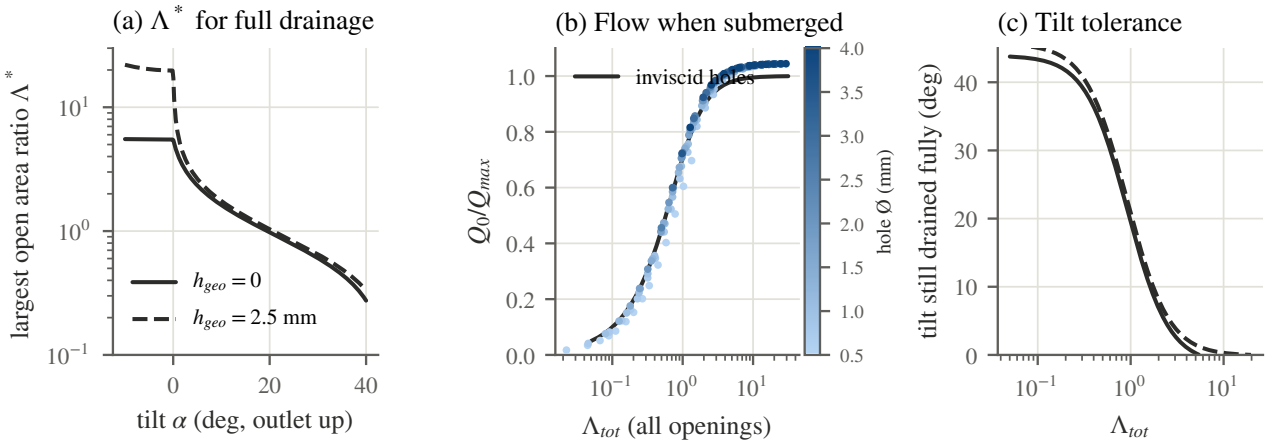


Figure 13: Closed-form design charts. (a) Largest exposed-area ratio that still drains to the lowest hole, Eq. (19). (b) Flow when submerged against total area ratio (line: inviscid holes in series with the bore; dots: network results, coloured by hole size). (c) Largest outlet-up tilt that still drains fully, if all holes are exposed at the end.

Table 4: Recommended layout (two staggered rows at ±30°) at about 28 mm^2 for each hole size: stop level h_s (mm).

Ø (mm)	holes	A (mm^2)	Q_0 (mL/min)	−10°	−5°	0°	+5°	+10°	+15°	+20°
0.5	144	28	699	1.7	1.2	0.6	1.0	1.4	1.9	2.3
1	36	28	755	2.0	1.3	0.5	0.9	1.4	1.8	2.1
1.5	16	28	773	2.6	1.6	0.5	0.9	1.3	1.7	2.0
2	8	25	747	3.7	2.1	0.4	0.8	1.2	1.6	1.9
2.5	6	29	794	4.4	2.4	0.3	0.8	1.2	1.5	1.8
3	4	28	786	5.9	3.2	0.3	0.7	1.1	1.4	1.7
3.5	2	19	670	10.6	5.5	0.3	0.7	1.0	1.4	1.6
4	2	25	756	10.5	5.4	0.2	0.6	1.0	1.3	1.5

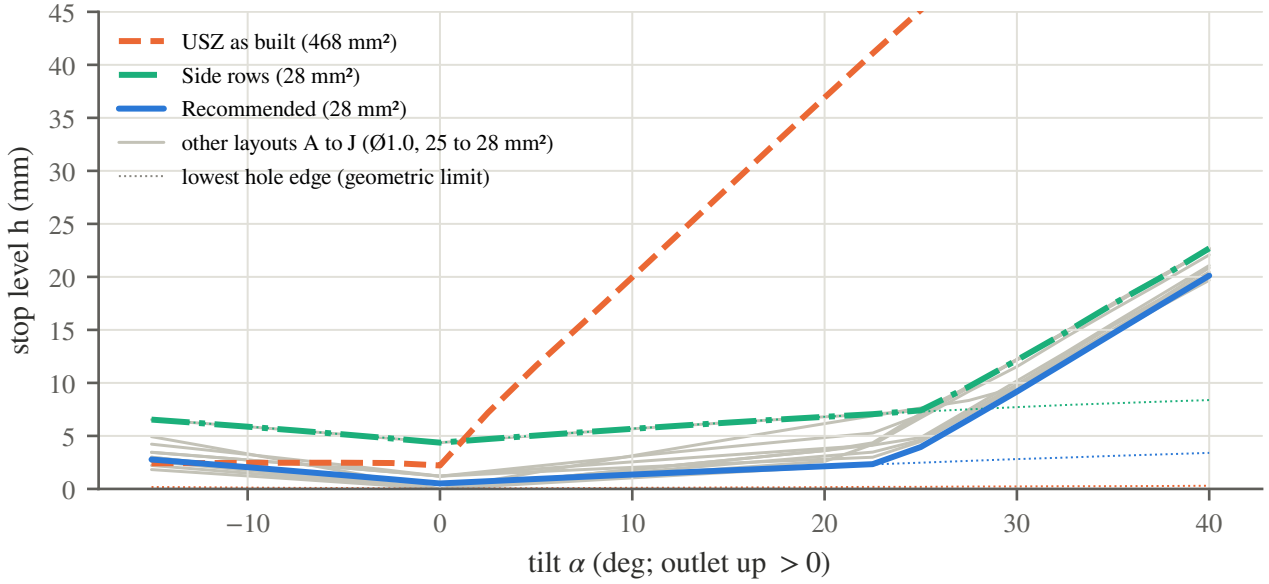


Figure 14: Stop level against tilt from −15° to +40°. Dotted lines mark the lowest hole edge, i.e. the best any layout can do.

6.5 When the source has limited capacity

The results so far assume a regulator that holds −7 mmHg whatever the air flow. Once the recommended drain is fully exposed it draws about 19 to 23 L/min of air, and the USZ field about 28 L/min. For the recommended layout, the smallest source capacity that still drains to the lowest hole is 9, 12, 15 and 18 L/min at +5°, +10°, +15° and +20°. This is the free-air flow a wall regulator must pass while holding −7 mmHg. If the source moves less than that, its vacuum collapses while holes are exposed, and with the outlet up the drain stops early (Fig. 15, Table 5). Equation (20) gives a conservative estimate of the allowed open area. A 10 L/min source allows about 16 to 19 mm^2, and a 5 L/min source about 9 mm^2. A layout fitted to the source capacity drains to the bottom again; when lying flat (z_ℓ = 2.7 mm) the source capacity makes no difference.

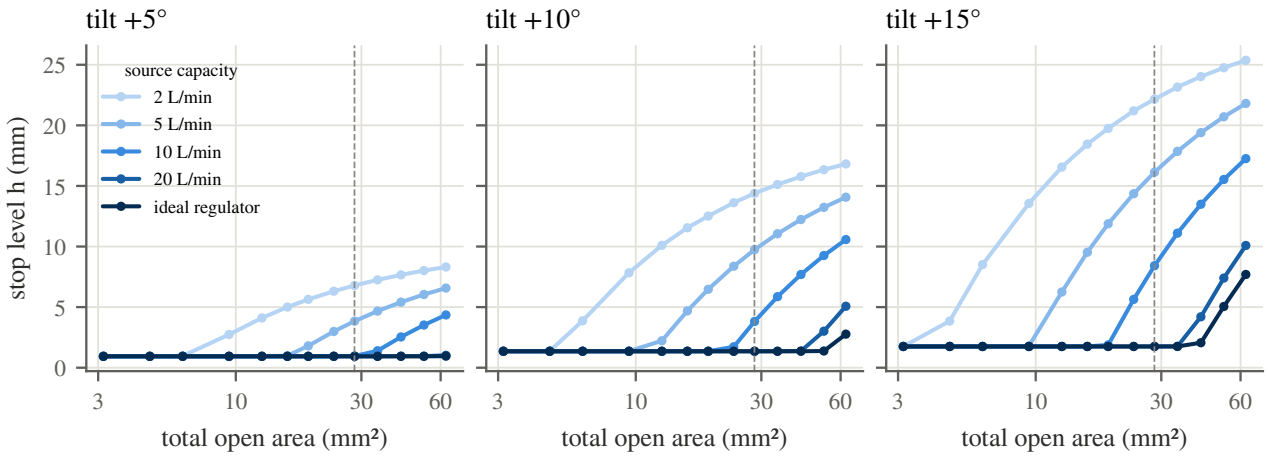


Figure 15: Stop level against total open area (recommended layout, Ø1.0 mm) for sources of limited flow capacity. The dashed line is twice the bore area.

Table 5: Largest total open area (recommended layout, Ø1.0 mm) that still drains to the lowest hole, against source capacity: network simulation and the conservative estimate of Eq. (20).

source capacity	largest open area that still drains fully (mm^2)		Eq. (20) (mm^2)	
	α = +10°	α = +15°	+10°	+15°
2 L/min	5	3	3	2
5 L/min	9	9	7	6
10 L/min	19	16	14	11
20 L/min	42	35	27	23
ideal regulator	52	35	–	–

6.6 Sensitivity and robustness

Table 6 and Fig. 16 vary one assumption at a time:

- **Liquid and suction level.** Viscosity (water to 80 % glycerol), suction (−5 to −10 mmHg) and the discharge coefficients change the flow rates but hardly the stop levels, because the stop is set by gas flow and hydrostatics.
- **Capillary holding.** It does not change the recommended design, which already drains fully, but it would help the as-built USZ field a lot (20 down to 3.6 mm at 10° with perfect wetting), because many small wetted holes can hold back air once the vacuum has dropped. The tube material and surface state matter here.
- **Drawdown.** Surface drawdown above active holes adds about half a hole diameter to every stop level.
- **Lying on a floor.** This blocks the bottom row of layout C, which then stops at mid-height (layout J would be blocked entirely). The recommended rows at ±30° are not affected.

In a Monte Carlo study (200 random combinations of tilt from −10° to +15°, suction, liquid, discharge coefficients, contact angle and drawdown; Fig. 17), the recommended design had a median stop level of 1.6 mm and a 95th percentile of 2.2 mm. For the as-built USZ field these were 4.8 mm and 24.6 mm.

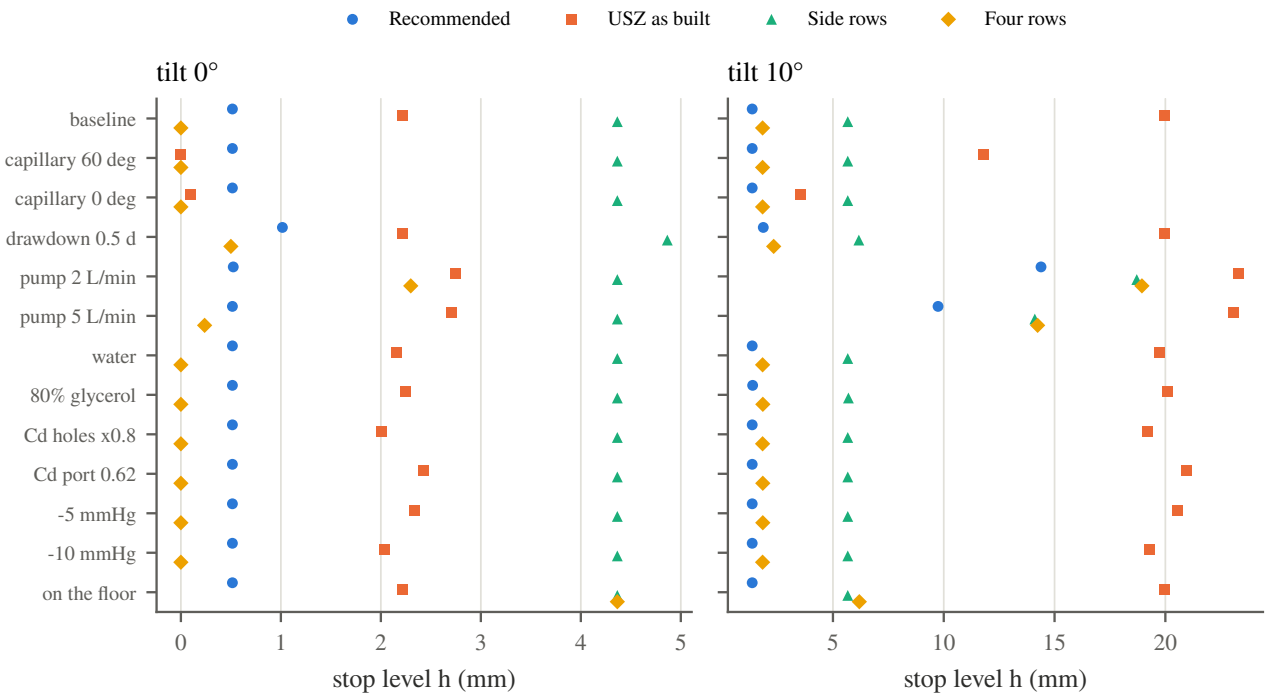


Figure 16: One-at-a-time sensitivity of the stop level, flat (left) and outlet 10° up (right).

Table 6: Sensitivity: stop level h_s (mm) at 0° and +10°.

case	Recommended		USZ as built		Side rows		Four rows	
	0°	+10°	0°	+10°	0°	+10°	0°	+10°
baseline	0.5	1.4	2.2	20.0	4.4	5.7	0.0	1.8
capillary 60 deg	0.5	1.4	0.0	11.8	4.4	5.7	0.0	1.8
capillary 0 deg	0.5	1.4	0.1	3.6	4.4	5.7	0.0	1.8
drawdown 0.5 d	1.0	1.9	2.2	20.0	4.9	6.2	0.5	2.3
pump 2 L/min	0.5	14.4	2.7	23.3	4.4	18.7	2.3	18.9
pump 5 L/min	0.5	9.7	2.7	23.1	4.4	14.1	0.2	14.2
water	0.5	1.4	2.2	19.7	4.4	5.7	0.0	1.8
80% glycerol	0.5	1.4	2.2	20.1	4.4	5.7	0.0	1.8
Cd holes x0.8	0.5	1.4	2.0	19.2	4.4	5.7	0.0	1.8
Cd port 0.62	0.5	1.4	2.4	20.9	4.4	5.7	0.0	1.8
-5 mmHg	0.5	1.4	2.3	20.5	4.4	5.7	0.0	1.8
-10 mmHg	0.5	1.4	2.0	19.3	4.4	5.7	0.0	1.8
on the floor	0.5	1.4	2.2	19.9	4.4	5.7	4.4	6.2

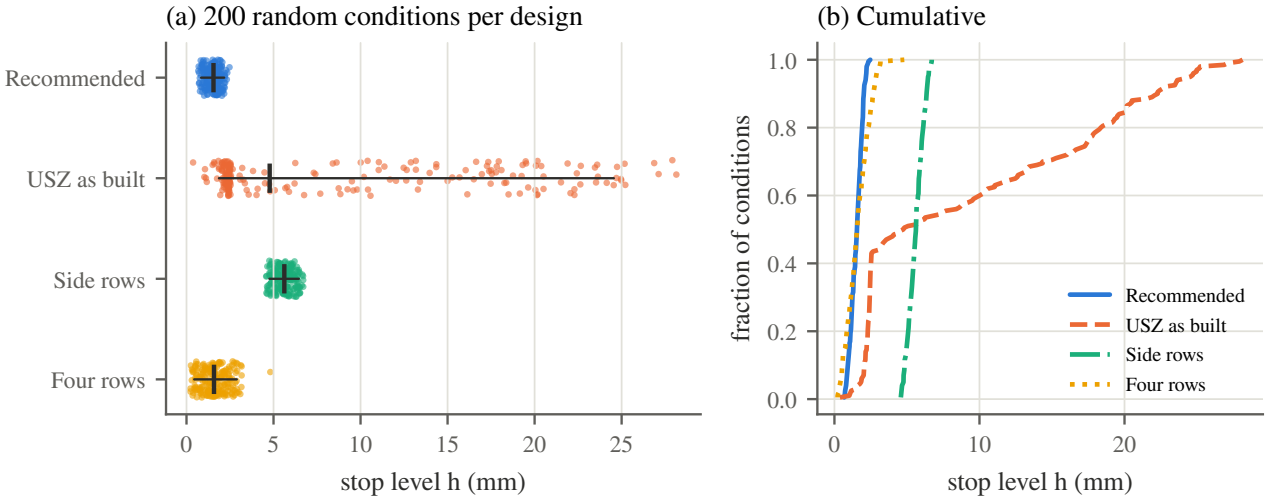


Figure 17: Monte Carlo study: stop level under 200 random operating conditions per design. (a) Each dot is one condition; the bars show the median and the 5th to 95th percentiles. (b) Cumulative distributions.

7 Recommended design

Table 7: Recommended hole layout for the USZ body at −7 mmHg.

rows	two rows at $\theta = \pm 30°$ from the bottom line (about 5 and 7 o'clock when lying flat); none on the upper half
alignment	staggered by half a pitch
holes	Ø1.0 mm (1.0 to 1.5 mm is equally good), 18 per row, pitch 6.0 mm
span	$x = 5$ to 113 mm, reaching as close to both ends as the part allows
open area	28.3 mm^2 ≈ 2× bore area ($\Lambda_{tot} = 1.44$)
suction source	wall regulator holding −7 mmHg while passing 18 L/min of air (12 L/min up to 10° tilt); with a 5 to 10 L/min source, reduce to 12 to 24 holes (Table 5)
expected stop level	0.5 mm (flat) to 2.1 mm (20°), i.e. the lowest hole edge; full drainage up to 22.5°
flow when submerged	755 mL/min (81 % of the bore limit)

The design rules, in order of importance:

1. Keep the total open area near 1 to 2 times the bore area. More area adds almost no flow, because the bore already limits it, but it destroys the vacuum once holes are exposed.
2. Put the holes on the lower half of the skin, but not on the contact line.
3. Spread them uniformly along the whole length, using small holes so that there are enough stations to reach both ends, and stagger the rows.
4. Match the open area to the flow capacity of the suction source.

A flat drain may come to rest upside down. Mirroring the rows onto the top face would double the exposed area and cost tilt tolerance, so marking the orientation is the better fix.

Suggested bench test. In the glycerol–water tank:

- tilt the drain to 0°, 5°, 10° and 15° with the outlet up, and record the residual level;
- measure the vacuum in the outlet tube close to the drain, to confirm that the wall regulator really holds −7 mmHg while the drain sucks air (about 23 L/min);
- compare the as-built USZ part with a part drilled to Table 7.

The model predicts that at 10° the as-built part leaves about 20 mm and the recommended part about 1.4 mm.

8 Limitations

- **Model order.** The model is a reduced-order network, not a 3D volume-of-fluid simulation. Slug and churn flow in the bore, foaming, and unsteady sloshing are represented only by the separated-phase bore model.
- **Interior.** The interior is one open lumen; the flow distribution along it (a suction manifold [10]) matters only for flow rates. The fluted inner wall and core of the as-built part add axial resistance, which changes flow rates and drain times but not the stop levels, since no liquid flows at the stop.
- **Hole and bore laws.** The discharge coefficients and the additive Sampson–Poiseuille law are standard correlations, not fitted to this part. The stop level is insensitive to them (Table 6).
- **Capillarity and drawdown.** These are treated with simple bounds. The real contact angle of the drain material should be measured.
- **Tissue and clots.** Tissue contact and clotting are outside the model; the existing FEBio work in the project addresses tissue sealing a hole.
- **Suggested CFD check.** A 3D volume-of-fluid run (for example OpenFOAM `interFoam` on the Euler cluster, as already set up for the sponge study) of two cases, the as-built field and the recommended layout at 10°, would be the next check.

9 Conclusions

1. **Exposed holes always leak.** At −7 mmHg ($H_v = 84.5$ mm), every hole of 0.5 to 4 mm lets air in once it is exposed; capillary sealing would need holes below 0.29 mm.
2. **The stop condition.** The air leak and the outlet bore form a pressure divider. Drainage stops when $h + H_v/(1+\Lambda_w^2) = z_\ell$, or when no hole is left below the surface. Two independent simulations confirm this.
3. **Tilt is the critical case.** When the drain lies flat, every layout drains to just below the bore lip (2.7 mm). With the outlet up, the lip rises by $L \sin\alpha$, and only a small exposed-area ratio keeps the vacuum high enough to lift the liquid there.

4. **What decides the result.** Hole size matters only through the total open area and the coverage of the length. Too much area (as built: 468 mm^2) stops the drain at 20 mm with the outlet 10° up; too few, large holes leave the ends uncovered.
5. **Recommendation.** Use two staggered rows at ±30° with 36×Ø1.0 mm (2 times the bore area). This drains to the lowest hole edge from −10° to +22° while keeping 81 % of the maximum flow. With the intended wall regulator this needs at least 18 L/min of air flow at −7 mmHg; with a weaker source, use fewer holes (Table 5).

Reproduction. All code is in `drain/drainstop`: `drainstop/model.py` (geometry, hydraulics, network solver, closed form), `drainstop/transient.py` (time-dependent model), `scripts/run_study.py` (all studies, ∼10 min on 12 cores), `scripts/make_figures.py` and `scripts/make_tables.py`.

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